

AgriRadius Pro - User Guide

How to use the app - a quick guide for new users. No technical knowledge needed.

What is it?

AgriRadius Pro turns any place in South India into a complete farming report from free satellite and government data. Pick a location and it shows crops, soil, rainfall, live weather, market prices and livestock/allied-sector data - useful for sourcing, input planning and field decisions.

1. Opening the app

- Open the link in **Google Chrome** (works on desktop or phone). No installation.
- If it asks for a **password**, type the one you were given.
- The **first load can take ~30 seconds** if the app was idle - it is waking up. After that it is fast.

2. The screen at a glance

The **left sidebar** is where you control everything; the **main area** shows the map and the results.

- **Sidebar:** Search & location, Analysis input, map Layers, Compute quality, and a Service-health panel.
- **Main area:** the interactive map on top, and below it the **Analysis Results** with many tabs.

On a phone the sidebar folds into the ■ menu (top-left).

3. Choose what to analyse

In the sidebar, under **Input**:

- **Analysis Mode: Area (radius)** studies a circle around a point; **Point location** studies one exact spot.
- **Input Method:** type coordinates, or switch to **Map Click** to just click on the map.
- You can also paste a **Google Maps link** or a place name in the Search box at the top and press **Search Place / Set from Google Location**.
- **Radius (km):** use the slider or type an exact value. Keep it modest (7-10 km) - very large circles are slow and use more compute.
- Pick the **Year** for the satellite analysis.

Example: paste 10 . 6588, 77 . 0089 (Pollachi), radius 10 km.

4. Reading the map

- In the sidebar **Layers** panel, tick a layer to turn it on. Each computes live the first time (10-30s), then loads instantly.
- **Overlay opacity** slider makes layers see-through so you can see the ground below.
- **Compute quality** (Light / Balanced / Heavy): Heavy = sharpest but heaviest; drop to **Light** if things are slow or the satellite service is busy.
- If tiles look missing after zooming, click **Refresh map** above the map.

Layer	Shows
Dynamic World	AI land cover (water, trees, crops...)
Cropland Confidence	Where two datasets agree it is farmland
Paddy / Maize	Detected rice / maize fields
Plantations / Banana	Coconut-arecanut / banana
Aquaculture ponds	Fish / prawn / farm ponds
Soil pH / OC / N	Soil health, painted on the map

5. Reading the results (the tabs)

Below the map, the **Analysis Results** section has tabs. Some fetch data only when you click a button (so the page never hangs).

Tab	What you get
Summary	Land-cover breakdown + confidence & stability
Villages	Per-village cropland, ranking, and the trained classifier
Charts	Visual breakdown of the area
Crop Cycle	Sowing-peak-harvest pattern; paddy & plantation checks
Rainfall	10-year rainfall history
Forecast	Live conditions now (rain, temp, soil, drying) + 16-day outlook
Soil	pH, organic carbon, nitrogen, texture
Allied Sectors	Livestock, dairy, feed, aquaculture, sericulture, fisheries
Mandi	Today's prices, MSP floor, price trend, variety grades
Ground Truth	Log what a field actually grows (improves accuracy)
Downloads	Build a full PDF / Excel report of everything

Buttons like **Get Prices**, **Show price trend**, **Detect Paddy**, **Measure pond area** only run when you click them - so nothing loads that you did not ask for.

6. How much to trust the numbers

- Open the **Data & Confidence** box at the top of the results for a plain-English guide.
- **Measured** things (rainfall, crop vigour, radar detection, prices) are reliable. **Modelled/classified** things (land-cover class, soil at 250 m) are best read as **ranges**, not exact.
- Rule of thumb: **trust the direction and the ranges; verify the edges on the ground**. Accuracy improves as your team logs ground truth.

7. Adding field observations

In the **Ground Truth** tab, pick a village, tick the crops you actually saw, and save. This trains the app to be more accurate for your region over time. Soil-card values can be added too.

8. Downloading a report

In **Downloads**, click **Build Full PDF/Excel Report** to get a complete document of the area - crops, soil, rainfall, allied sectors and (if you fetched them) prices. Good for sharing offline.

9. Please test mindfully

This runs on a **free, open-source** setup (Google Earth Engine) with a **limited shared monthly compute budget**. To keep it fast for everyone:

- Enable **one heavy layer at a time**; turn others off.
- Keep the **radius reasonable** (7-10 km).
- Avoid **rapid repeated clicks**; let a layer finish.
- Use **Light** compute quality if it feels slow.
- The sidebar **Service health** panel shows how much monthly compute is left (it resets on the 1st).

Quick troubleshooting

- **Slow first open?** The app was asleep - wait ~30s.
- **A layer says 'Earth Engine is busy'?** Wait a minute, click Refresh, or switch to Light quality.
- **Tiles missing after zoom?** Click **Refresh map**.
- **Map jumped?** Re-search your location.

OneRoot AgriRadius Pro - Agricultural GIS Intelligence. Data: Sentinel-1/2 & Dynamic World (ESA/Google), SoilGrids, CHIRPS, Open-Meteo, AGMARKNET (data.gov.in), Livestock Census 2019.